

PROPOSED SYSTEM PLAN

Optimizing Construction Logistics Using QR Code Inventory Management for Material Tracking in Bulalacao, Oriental Mindoro

System Users, Modules, Functional Scope, and Laravel Development Framework

1. Purpose of the Proposed System

The proposed system is a QR Code-based Construction Inventory Management System designed to improve the tracking, monitoring, and control of construction materials in Bulalacao, Oriental Mindoro. The system will replace or reduce dependence on manual material recording by allowing authorized personnel to register materials, assign QR codes, scan materials, record material movements, update inventory, monitor locations, and generate reports.

The system is specifically aligned with the study's objectives of improving material tracking accuracy, time efficiency, and inventory control.

2. Identified System Users

User	Primary Purpose	Main Responsibilities
Administrator	Manage the overall system and user access.	Manage user accounts, roles, permissions, system/master data, activity logs, and access to reports.
Inventory Officer / Inventory Staff	Manage construction material records and inventory.	Register materials, receive materials, generate QR codes, update inventory, record transactions, monitor stock, and prepare reports.
Site Personnel / Material Handler	Record the physical movement and handling of materials.	Scan QR codes, view material information, record movement, transfer materials, record usage/returns, and report damaged or missing materials.
Project Manager / Supervisor	Monitor inventory and construction material logistics.	View dashboard information, monitor stock and material movement, review losses/damages, and access management reports.

Note: Suppliers are not treated as direct system users unless the research scope later requires supplier access. Supplier information may instead be maintained by the Inventory Officer.

3. Proposed System Modules

The proposed system will consist of eight (8) major modules. The items listed under each module are functional features/sub-functions rather than separate stand-alone modules.

No.	Main Module	Functional Features
1	1. Dashboard	• Inventory Summary • Material Movement Summary • Low Stock Alerts • Recent Transactions
2	2. User and Access Management	• User Management • Role Management • Permission Management • User Activity Logs
3	3. Material Management	• Material Registration • Material Categories

		• Material Information • Stock Level Management • Supplier Information • Material Status
4	4. QR Code Management	• QR Code Generation • QR Code Assignment • QR Code Printing • QR Code Scanning • QR Code Validation
5	5. Material Transactions	• Material Receiving • Material Issuance • Material Transfer • Material Usage • Material Return • Damaged Material Recording • Lost/Missing Material Recording
6	6. Location Management	• Construction Site Management • Storage Area Management • Material Location Assignment • Material Transfer History
7	7. Inventory Monitoring	• Current Inventory • Stock-In Monitoring • Stock-Out Monitoring • Low Stock Monitoring • Damaged/Lost Material Monitoring • Inventory History
8	8. Reports and Analytics	• Inventory Report • Material Movement Report • Receiving Report • Issuance Report • Transfer Report • Damaged/Lost Material Report • Low Stock Report • User Activity Report • QR Scanning/Transaction History

4. Module and Feature Summary

No.	Main Module	Number of Functional Features
1	1. Dashboard	4
2	2. User and Access Management	4
3	3. Material Management	6
4	4. QR Code Management	5
5	5. Material Transactions	7
6	6. Location Management	4
7	7. Inventory Monitoring	6
8	8. Reports and Analytics	9
	TOTAL	45

Final scope: 8 major modules and 45 functional features. The 45 items should be presented in the documentation as functions/features within the eight major modules, not as 45 independent modules.

5. Core System Workflow

1. Material Registration
2. QR Code Generation
3. QR Code Assignment and Printing
4. QR Code Tagging of Material or Material Batch

5. QR Code Scanning
6. Material Transaction Recording
7. Location and Quantity Updating
8. Real-Time Inventory Update
9. Material Movement History
10. Dashboard Monitoring
11. Report Generation

Example: When 50 bags of cement are received, the Inventory Officer registers the material and assigns a unique QR code. When the material is stored, issued, transferred, used, returned, damaged, or reported missing, the authorized user scans the QR code and records the corresponding transaction. The system then updates the inventory and preserves the transaction history.

6. Material Transaction Types

Transaction Type	Purpose
Received	Records materials entering the construction inventory.
Issued	Records materials released from inventory for a project or activity.
Transferred	Records movement of materials from one location to another.
Used	Records materials consumed during construction activities.
Returned	Records unused materials returned to storage.
Damaged	Records materials that can no longer be used normally.
Lost/Missing	Records materials that cannot be located or accounted for.

7. Development Framework and Technology

Technology	Proposed Use
Laravel Framework	Primary backend web application framework using the MVC architecture.
PHP	Server-side programming language used by Laravel.
MySQL	Relational database for users, materials, QR references, locations, transactions, and inventory records.
Blade Templates	Laravel's server-side templating system for web interfaces.
HTML/CSS	Structure and styling of the system interface.
JavaScript	Client-side interactions, dynamic forms, and QR scanning interface where applicable.
QR Code Technology	Identification and quick retrieval of material records.
Eloquent ORM	Laravel's database abstraction layer for interacting with MySQL.
Laravel Authentication/Authorization	User login, role-based access, and permission control.

8. Proposed Laravel MVC Architecture

User / QR Scanner / Web Browser

Laravel Routes

Laravel Controllers

Laravel Models and Eloquent ORM

MySQL Database

Laravel Views / Blade Interface

The general flow is: User → Laravel View/Interface → Route → Controller → Model/Eloquent → MySQL Database → Controller → View/Interface.

9. Suggested Database Tables

Table	Purpose
users	Stores system user accounts and basic user information.
roles	Stores user roles.
permissions	Stores access permissions, if role-based permissions are implemented.
activity_logs	Records important user/system activities.
materials	Stores material information and current inventory-related attributes.
material_categories	Stores material categories.
suppliers	Stores supplier information.
qr_codes	Stores QR references associated with materials or material batches.
transactions	Stores all material movement and inventory transactions.
sites	Stores construction site information.
locations	Stores warehouses, storage areas, and other material locations.
inventory	Stores current inventory balances, if maintained separately from material master data.
inventory_adjustments	Stores authorized inventory corrections or adjustments.

The final database structure may be adjusted during system analysis and ERD design. Tables should not be added simply to increase the number of tables; each table should have a clear relationship to an actual system requirement.

10. Alignment with the Statement of the Problem

Research Question	System Components That Address It
1. What are the key limitations of current material tracking methods in construction?	Analysis of current/manual practices, material handling records, historical losses, and existing inventory tools.
2. How can a QR code system be designed to streamline material logging and retrieval?	QR Code Management, Material Management, Material Transactions, and Location Management.
3. How effective is the proposed system in improving tracking accuracy, time efficiency, and inventory control?	QR scanning records, Inventory Monitoring, Reports and Analytics, time-motion comparison, accuracy checking, and user evaluation.

11. Proposed System Evaluation

To determine whether the proposed system improves construction material tracking, the study may compare the existing/manual process with the QR-based system using measurable indicators.

Tracking Accuracy – compare recorded inventory with actual physical inventory.

Time Efficiency – compare the time required for material logging, retrieval, and transaction recording.

Inventory Control – evaluate the ability to monitor stock levels, movement, losses, and discrepancies.

Usability – evaluate whether intended users can understand and operate the system.

User Satisfaction – obtain feedback from authorized users regarding the system's usefulness and ease of use.

12. Alignment with the IPO Conceptual Framework

INPUT

- Demographic profile of site personnel and material handlers
- Current material handling and inventory practices
- Historical material loss and handling inefficiency records
- Existing inventory control and logging tools
- Material, user, location, and inventory requirements

PROCESS

- Requirements gathering and analysis
- System design and Laravel development
- Database development
- QR code generation and tagging
- Material scanning and transaction recording
- Site testing and time-motion study
- Relative Importance Index (RII) analysis of identified handling issues
- Statistical evaluation of handling time and inventory accuracy
- User acceptance and system evaluation

OUTPUT

- QR Code Inventory Management System
- Real-time material tracking
- Improved inventory accuracy
- Faster material logging and retrieval
- Improved monitoring of material movement
- Reduced material discrepancies and improved accountability
- Standardized digital material-handling procedure

- Inventory and material movement reports
- Recommendations for digital material-handling integration

13. Recommended Final Scope for the Thesis

For consistency throughout the proposal, system documentation, ERD, use-case diagram, prototype, implementation, and evaluation, the recommended final scope is four (4) user types and eight (8) major modules. The 45 listed functional features are grouped under these eight modules. Laravel will serve as the primary web application framework, with MySQL as the relational database.

The central function of the system is not only inventory counting. It is the digital tracking of construction material movement through QR identification, transaction recording, location monitoring, and inventory updating.